

Erythroderma (exfoliative dermatitis). Part 1: underlying causes, clinical presentation and pathogenesis

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CPD

Erythroderma (exfoliative dermatitis). Part 1: underlying causes, clinical presentation and pathogenesis

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Summary

Erythroderma (exfoliative dermatitis), first described by Von Hebra in 1868, manifests as a cutaneous inflammatory state, with associated skin barrier and metabolic dysfunctions. The annual incidence of erythroderma is estimated to be 1–2 per 100 000 population in Europe with a male preponderance. Erythroderma may present at birth, or may develop acutely or insidiously (due to progression of an underlying primary pathology, including malignancy). Although there is a broad range of diseases that associate with erythroderma, the vast majority of cases result from pre-existing and chronic dermatoses. In the first part of this two-part concise review, we explore the underlying causes, clinical presentation, pathogenesis and investigation of erythroderma, and suggest potential treatment targets for erythroderma with unknown causes.

Introduction

Erythroderma (exfoliative dermatitis) is a dermatological emergency, and presents with extensive erythematous skin and scaling, affecting $\geq 90\%$ of body surface area. First described by Von Hebra in 1868, erythroderma represents a cutaneous inflammatory state, with associated dysfunction of both the skin barrier and metabolic processes. Many diseases are associated with erythroderma, and most cases result from pre-existing and chronic dermatoses.¹ In Part 1 of this concise review, we explore the underlying causes, clinical presentation, pathogenesis and investigation of

erythroderma, and suggest potential treatment targets for erythroderma of unknown cause.

Underlying causes of erythroderma

Erythroderma is the result of severe skin dysmetabolism, reported in severe forms of a range of diseases, which can be broadly categorized into congenital, infective, inflammatory, immunobullous, neoplastic, iatrogenic and idiopathic causes. Table 1 shows the range of diseases associated with the clinical presentation of erythroderma, including the key clinical features and useful references for further reading. Commonly, erythroderma results from exacerbations of pre-existing dermatoses, such as psoriasis or eczema. Psoriatic erythroderma may result from sudden withdrawal of systemic or very potent topical corticosteroids. Drug-related erythroderma eruptions are often caused by anticonvulsants, antibiotics and topical

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Table 1 Diseases associated with erythroderma and useful diagnostic features.

Category	Examples	Useful diagnostic features and investigations	Reading list of useful references
Congenital	Ichthyosis and keratinopathies	Clinical history	Hoeger and Harper, 1998; ¹⁷ Dhar et al., 2012; ¹⁸ Louhichi et al., 2019 ¹⁹
	Syndromic forms without skin blisters, such as:	Starts at birth. Inherited forms of acrodermatitis enteropathica may commence after the infant weans	
	Netherton syndrome (AR)	Clinical signs	
	Sjögren–Larsson syndrome (AR)	Collodion membrane may be present at birth in syndromic and nonsyndromic forms of congenital ichthyosis	
	Dorfman–Chanarin syndrome (AR)		
	Refsum syndrome (AR)		
	KID syndrome (AR)		
	Conradi–Hünermann–Happle syndrome (XLD)	Recessive X-linked ichthyosis affects only males, and typically presents with generalized scale within 6 months of birth. May co-present with undescended testicles and corneal opacities	
	CHILD syndrome (XLD)		
	IFAP syndrome (XLR and AD)		
Inflammatory	X-linked ichthyosis (XLR)		Akhyani et al., 2005; ²⁰ Mathew and Sreedevan, 2017 ²¹
	Nonsyndromic forms, such as:	Investigations	
	Lamellar ichthyosis (AR)	Relevant gene panels	
	Nonbullous congenital ichthyosiform erythroderma (AR)	Serum immunoglobulin level in Omenn syndrome	
	Harlequin ichthyosis (AR)	Serum zinc level in inherited forms of acrodermatitis enteropathica	
	Nonsyndromic forms with blistering such as:		
	Epidermolytic ichthyosis (AD and AR)		
	Other causes, such as:		
	Immunodeficiency syndromes		
	Omenn syndrome (AR)		
	Hyper IgE syndrome		
	Metabolic disorders		
	Holocarboxylase synthetase deficiency (AR)		
	Inherited form of acrodermatitis enteropathica (AR)		
	Eczema	Clinical history	
	Psoriasis	Associated atopy; family history of atopy or psoriasis; previous clinical diagnosis of dermatosis. History of dystrophic nails and inflammatory arthropathy may favour a diagnosis of psoriasis	
	PRP	PRP typically presents with palmoplantar keratoderma	
	Seborrhoeic dermatitis	Clinical signs	
		Palmoplantar keratoderma in PRP	
		Nail pitting and dystrophy in psoriasis	
		Investigations	
		Full blood count (especially eosinophil count) and skin biopsy	
		HIV screening in severe seborrhoeic dermatitis	
		Total immunoglobulin E level in atopic dermatitis	

Table 1 continued

Category	Examples	Useful diagnostic features and investigations	Reading list of useful references
Immunobullous	Subcorneal: PF Intraepidermal: PV Subepidermal: BP	Clinical history Patients may not report any blisters in PV and PF. History of oral and genital ulceration is a key history in PV BP may present with a prebullous stage with widespread urticated pruritic plaques before blistering commences Clinical signs Scale, crust, erosion, typically confined to the skin in PF but sparing the mucosa Mucosal erosions and ulceration are a key feature in PV, and may or may not affect the skin Urticated plaques, larger intact and deroofted bullae in BP Investigations Skin histology: PF and PV show intraepidermal blistering; BP shows subepidermal blistering Skin direct IMF: PV and PF shows intercellular surface staining pattern; BP shows linear basement membrane zone pattern ELISA: key features are Dsg1 in PF; Dsg3 in PV; BP180 and BP230 in BP	Scrivener et al., 1998 ²²
Infective	HIV Scabies SSSS	Clinical history and signs Opportunistic and recurrent infections in HIV Widespread pruritus and erythematous rash with burrows on dermoscopy in scabies Fever, malaise and widespread fluid-filled blistering rash in SSSS	Mathew and Sreedevan, 2017; ²¹ Bowles and Smimov, 2019; ²³ Papanizos et al., 2019; ²⁴ Lim et al., 2019 ²⁵
Neoplastic	Sézary syndrome Cutaneous T-cell lymphoma B-cell lymphoma Paraneoplastic pemphigus Malignancies, particularly haematological and solid organ	Clinical history and signs Unexplained weight loss, lymphadenopathy, pruritus Investigations Blood test screening Flow cytometry Skin biopsy Lymph node/bone marrow biopsy	Lim et al., 2018 ²⁵

Table 1 continued

Category	Examples	Useful diagnostic features and investigations	Reading list of useful references
Immune	Hypersensitivity	Clinical history and signs	
	Contact dermatitis	EMM: about 90% cases caused by infections (particularly herpesvirus) and 10% caused by drugs; presenting as target-like erythematous lesions and polymorphous rash	
	EMM		
	Steven-Johnson syndrome (90% cases)		
Immunodeficiency	Immunodeficiency	SJS: 90% cases associated with infections and typically presents with mucocutaneous skin erosions and conjunctivitis	
	GvHD	History of allogeneic transplant in GvHD. Acute GvHD occurs within 3 months of transplantation and may be associated with gastrointestinal and liver dysfunction. Chronic GvHD may associate with dry eyes, scleroderma, hair loss, nail dystrophy, liver, lung and gastrointestinal dysfunction	
	Autoimmune	Extracutaneous features in SLE can include arthritis, pleurisy, pericarditis, kidney disease, neuropathy, etc.	
	SLE		
Iatrogenic		Investigations	
		Blood test screening – autoantibodies.	
		Skin biopsy. Patch testing in contact dermatitis	
	Drugs such as antibiotics, anticonvulsants, antimalarials, allopurinol, lithium, barbiturates	Clinical history	
EMM (10% cases)	EMM (10% cases)	May present with systemic symptoms such as malaise, fever	
	DRESS	Clinical signs	
	Steven-Johnson syndrome (10% cases)	Lymphadenopathy and hepatosplenomegaly may present in DRESS syndrome. Pustular lesions developing at flexural sites may suggest AGEF or bacterial skin infection	
	Toxic epidermal necrolysis		
AGEP	AGEP	Investigations	
		Blood test screening for eosinophilia and deranged liver function test.	
		Skin biopsy	
		Clinical history	
Nutritional	Acrodermatitis enteropathica (inherited or acquired)	Poor nutrition, excess alcohol, history of malabsorption	Hoeger and Harper, 1998 ¹⁷
		Clinical signs and symptoms	
		Triad of psoriasiform dermatitis of circumoral or periorificial areas, alopecia and diarrhoea	
		Investigations	
		Blood test screening for nutritional deficiencies	

Table 1 continued

Category	Examples	Useful diagnostic features and investigations	Reading list of useful references
Others	Cutaneous mastocytosis Hypereosinophilic syndrome Sunburn	Clinical history and sign Mastocytosis: rubbing an area of affected skin can activate mast cells leading to skin inflammation (Darier sign). Investigations Skin biopsy and haematological assessment in mastocytosis and hypereosinophilic syndrome	AGEP, acute generalized exanthematous pustulosis; AR, autosomal recessive; AD, autosomal dominant; BP, bullous pemphigoid; CHILD, congenital hemidysplasia with ichthyosiform erythroderma and limb defects; DRESS, drug reaction with eosinophilia and systemic symptoms; Dsg, desmoglein; EMM, erythema multiforme major; GvHD, graft-versus-host disease; HIV, human immunodeficiency syndrome; IFAP, ichthyosis follicularis, atrichia and photophobia; KID, keratitis-ichthyosis-deafness; IMF, immunofluorescence microscopy; PF, pemphigus foliaceus; PRP, pityriasis rubra pilaris; PV, pemphigus vulgaris; SJS, Steven-Johnson syndrome; SLE, systemic lupus erythematosus; SSSS, <i>Staphylococcus</i> scalded skin syndrome; XLD, X-linked dominant; XLR, X-linked recessive.

preparations.² A detailed chronological history, including clinical improvement following drug cessation, are essential for the determination of the causative drug, which can often prove challenging.² Table 2 shows the distribution of the various causes of erythroderma. Despite the current knowledge and understanding of the many and varied causes of erythroderma, the precise cause may not be established in up to one in six cases.²

Clinical presentation of erythroderma

Erythroderma may present at birth, and develops either acutely (e.g. due to infection or drugs), or more gradually due to the progression of an underlying primary pathology, including dermatoses and malignancies.² Associated signs and symptoms may give further clues to the underlying aetiology of erythroderma.

Congenital

Congenital erythroderma may co-present with a diversity of cutaneous and extracutaneous clinical features at birth and later in life. In the case of autosomal recessive congenital ichthyosis, affected infants may co-present with collodion membrane, ectropion, eclabium and folded ears at birth, which may be further complicated, in the case of harlequin ichthyosis, by hair and nail abnormalities, hypohidrosis and skin contractures. In Netherton syndrome, the skin has the characteristic 'double-edged' ichthyosis linearis circumflexa appearance, the hair shaft is abnormal and brittle (trichorrhexis invaginata), and affected children have a predisposition to atopy. Developmental delays in children born with congenital erythroderma should always prompt the screening of syndromic forms of congenital ichthyosis and neutral lipid storage disease (e.g. Charnarin-Dorfman syndrome). In erythroderma due to primary acrodermatitis enteropathica, the presentation of perioral and perianal dermatitis may not present until after the affected infant begins to wean off breast milk. Congenital erythroderma may also co-present in severe combined immunodeficiency, a rare disease associated with alopecia, chronic diarrhoea, failure to thrive, lymphadenopathy and hepatosplenomegaly.

Acute

Acute erythroderma usually manifests clinically as widespread erythema, followed by exfoliative scaling over the subsequent 2–6 days.³ The morphology of the

Table 2 Literature review of the differential causes of erythroderma from different geographical locations worldwide.

Reference	Description of study	Differential causes	Demographics (age and sex)
Akhyani <i>et al.</i> , 2005 ²⁰	Case series of 97 patients with acquired erythroderma in Iran	Dermatoses 59.7% Drug reactions 21.6% Malignancies 11.3% Idiopathic 7.2%	Median age at diagnosis 46.2 years; male : female ratio 1.85 : 1
Leenutaphong <i>et al.</i> , 1999 ²⁶	Case series of 49 patients with acquired erythroderma in Thailand	Drugs 38.77% Dermatoses 27.5%	Median age at presentation 51.7 years; male : female ratio 2 : 1
Miyashiro and Sanches, 2020 ²	Case series of 309 patients in Brazil	Eczema 20.7% Psoriasis 16.8% Sézary syndrome 12.3% Drug eruption 12.3% Atopic dermatitis 8.7% Mycosis fungoides 5.5% Others 6.8% Idiopathic 16.8%	Median age 57 years; male : female ratio 2.2
Sudho <i>et al.</i> , 2003 ²⁷	Case series of 25 patients in Porur, India	Psoriasis 32% Drugs 24% Eczema 12% Hereditary disorders 8% PF 4% Other 12% ^a	Age range of 5 days to 72 years; peak incidence 21–30 years (24%); male : female ratio 1.5 : 1
Mathew and Sreedevan, 2017 ²¹	Case series of 370 patients in Kerala, India	Psoriasis 32.7% Contact dermatitis 15.9% Idiopathic 15.7% Chronic actinic dermatitis 8.1% Drugs 6.5% Atopic dermatitis 6.5% Malignancy 3.2% PRP 3.2% 1.4% PF Other 0.8% ^b	Mean age of onset of erythroderma 55.38 ± 16.67 years (range 3–91 years); male : female ratio of 3.6 : 1
Yuan <i>et al.</i> , 2010 ²⁸	Case series of 82 patients in China	Pre-existing 72% dermatoses (psoriasis 30.5%) Drug reactions 17% Idiopathic causes 6.1% Malignancies 4.9%	–
Rym <i>et al.</i> , 2005 ²⁹	Retrospective study of 80 erythrodermic adults in Africa	Psoriasis 51.25% Drugs 11.25% Malignancy 8.75% Idiopathic 7.5% PF 6.25% Contact dermatitis 2.5% PRP 1.25% PF 1.25%	Mean ± SD age 53.78 ± 18 years; male : female ratio 2.2 : 1.0
Pal and Haroon, 2002 ³⁰	Case series of 90 patients in Pakistan	Psoriasis 37.8% Idiopathic 14.6% CIE 7.8% PF 5.6% Drugs 5.5% Malignancy 5.5% Atopic dermatitis 3.3% Contact dermatitis 3.3% Crusted scabies 2.2% PRP 2.2%	Mean age of onset 41.6 years; male : female ratio of 2.8 : 1

CIE, congenital ichthyosiform erythroderma; PF, pemphigus foliaceus; PRP, pityriasis rubra pilaris; SSSS, *Staphylococcus* scalded skin syndrome. ^aIncluding malignancy, dermatophytosis, SSSS and PRP; ^bincluding crusted scabies and CIE.

rash preceding the erythroderma varies markedly, depending on the underlying cause, and particular attention to co-presenting symptoms is required to aid in the diagnosis of the underlying cause. Viral exanthems typically present as an acute morbilliform macular and/or maculopapular eruption spreading in a cephalocaudal direction. Drug eruptions may begin as a morbilliform or urticarial rash, which can further develop into pustules at flexural sites (in acute generalized exanthematous pustulosis), lymphadenopathy (in drug reaction with eosinophilia and systemic symptoms) or mucocutaneous skin erosions and blistering (in Stevens–Johnson syndrome and toxic epidermal necrolysis). Patients presenting with erythrodermic psoriasis may co-present with severe nail dystrophy. Erythroderma due to underlying immunobullous disease may present with intact and deroofed mucocutaneous blisters or erosions. Erythroderma in the context of unexplained weight loss, lymphadenopathy and presence of atypical lymphocytes (in blood film) in the absence of pre-existing skin diseases should prompt consideration of underlying malignancies such as Sézary syndrome.

Associations and sequelae

Thermoregulatory disturbances (fever and chills), reactive lymphadenopathy and superimposed bacterial infections may occur concurrently with erythroderma from all causes. In extreme cases, sequelae of erythroderma include hypovolaemia (with reflex tachycardia), high-output cardiac failure, anaemia, electrolyte disturbance and acute respiratory distress syndrome.^{3,4}

Pathogenesis of erythroderma

Congenital

Disorders of lipid metabolism underpin the pathogenesis of a range of differential causes of congenital erythroderma. Reviews by Elias *et al.*⁵ and Radner *et al.*⁶ extensively discuss the pathogenesis of genodermatoses secondary to abnormalities and dysregulation in fatty acid, cholesterol and triglyceride metabolism and in lipid transportation. In brief, disordered lipid metabolism leads to abnormal lipid organization within the stratum corneum, therefore interfering with the normal lateral packing of lipids, which is crucial for the barrier function of the skin.⁷ This abnormality has been observed in lamellar ichthyosis and atopic dermatitis.⁸

Acquired

The pathogenesis of acquired erythroderma is poorly understood. Studies suggest that factors such as an increase in serum IgE, interleukin (IL)-4 and IL-10, associated with T helper (Th)1/Th2 imbalance in favour of Th2 differentiation and dysregulated angiogenic factors could have a role in the pathogenesis of erythrodermic psoriasis.⁹ IL-4 and IL-13 signalling are central to the pathogenesis of atopic dermatitis and other atopic diseases through activation of downstream IgE-dependent processes, which lead to activation of mast cells and eosinophils and to the clinical presentation of skin inflammation.¹⁰ Indeed, the presence of dermal eosinophilic infiltrates is a characteristic histological feature observed in skin biopsies from patients with erythrodermic atopic dermatitis,¹¹ and this finding may also be observed in allergic contact dermatitis, urticaria, immunobullous diseases and dermal hypersensitivity reaction patterns.¹² IL-4 and IL-13 are targets for blockade by dupilumab, a biologic licensed for treatment of moderate to severe atopic dermatitis.

The pathogenesis of erythroderma in patients with haematological malignancies is poorly understood, and changes to the cytokine signalling profile in this group of patients are highly complex. Nevertheless, there is an increase in the Th2 profile of cytokines such as IL-4, IL-5 and IL-10 in the skin and serum of patients with Sézary syndrome,¹³ overexpression of IL-13 signalling in cutaneous T-cell lymphoma,¹⁴ and upregulation of IL-4 signalling in the bone marrow in patients with acute myeloid leukaemia.¹⁵ The link between IL-4 and IL-13 signalling in the pathogenesis of erythroderma in patients with haematological malignancies remains speculative, given the lack of relevant studies.

Final common pathway

Regardless of the underlying cause of erythroderma, there is an increase in expression of adhesion molecules (increased expression of vascular cell adhesion molecule-1, intercellular adhesion molecule-1, E- and P-selectin found on endothelium *in situ* in patients with erythroderma, mycosis fungoides and atopic dermatitis)¹⁶ and proinflammatory mediators stimulating dermal inflammation and epidermal turnover rate, which shortens the keratinocyte transit time through the epidermis, resulting in exfoliation.¹

Discussion

Our literature review identifies that globally, the most common cause of erythroderma is due to an underlying primary skin disease.² However, to our knowledge, there are no recent UK data about the differential causes of erythroderma. Therefore, advancement of the current clinical management of erythroderma is reliant upon developing current knowledge of the underlying pathogenesis in order to identify possible targets for therapeutic intervention. Identification of the specific molecular cytokines with raised levels in erythroderma may allow clinicians to subtype erythroderma cases and manage them accordingly. The pathogenesis of erythroderma remains unclear, but the IL-4 and IL-13 pathways appear to be an attractive potential treatment target for managing erythroderma with unclear cause.

Conclusion

A wide range of diseases can lead to erythroderma but its pathogenesis remains poorly understood.

Learning points

- Infants presenting with erythroderma at birth or within the first year of life should always prompt the consideration of inherited metabolic diseases.
- Pre-existing dermatoses (particularly psoriasis and eczema) underlie a significant proportion of cases of erythroderma, as shown in the current literature to date.
- Regarding the acute presentation of erythroderma, the morphological nature of the rash preceding erythroderma and co-presenting symptoms are of great significance to aid in the aetiological diagnosis.
- In erythrodermic psoriasis, studies have suggested factors such as increased serum IgE, IL-4 and IL-10, associated with favoured Th2 differentiation and dysregulated angiogenic factors, could have a role in the pathogenesis.

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CPD questions

Learning objective

To understand the underlying causes, clinical presentation, pathogenesis, investigations and potential treatments for erythroderma.

Question 1

A 45-year-old man presents with erythroderma secondary to a drug reaction to an antibiotic. How many days later would subsequent exfoliative scaling be expected to occur?

- (a) 0–1.
- (b) 2–6.
- (c) 10–14.
- (d) 14–28.
- (e) 28–56.

Question 2

After being weaned off breast milk, an infant develops perioral and perianal dermatitis, which gradually developed into erythroderma. The condition does not improve with regular emollients and moderate-potency corticosteroids. Which trace-element deficiency would be expected with the underlying disorder?

- (a) Low potassium level.

- (b) Low magnesium level.
- (c) Low calcium level.
- (d) Low sodium level.
- (e) Low zinc level.

Question 3

A 66-year-old man with Type 1 diabetes presents with a 12-day history of a widespread erythematous rash that started on his face and spread to affect most of his body, along with diffusely thickened and yellowish palms and soles. There were areas of uninvolved skin on his chest. What is the likely cause for his erythroderma?

- (a) Eczema.
- (b) Pityriasis rubra pilaris (PRP).
- (c) Psoriasis.
- (d) Scabies.
- (e) Sézary syndrome (SS).

Question 4

Which pathway appears to be an attractive potential treatment target for managing erythroderma with an unclear cause?

- (a) Tumour necrosis factor (TNF)- α .
- (b) Interleukin (IL)-17.
- (c) IL-4 and IL-13.
- (d) CD-20.
- (e) IL-12 and IL-23.

Question 5

A 5-day-old girl present widespread erythema and scaling covering 95% of her body with blistering and skin fragility. What is the most likely diagnosis?

- (a) Conradi–Hünemann–Happle syndrome.
- (b) Epidermolytic ichthyosis.
- (c) Lamellar ichthyosis.
- (d) Netherton syndrome.
- (e) Refsum syndrome.

Instructions for answering questions

This learning activity is freely available online at <http://www.wileyhealthlearning.com/ced>

Users are encouraged to

- Read the article in print or online, paying particular attention to the learning points and any author conflict of interest disclosures.
- Reflect on the article.
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- Complete the required evaluation component of the activity.

Once the test is passed, you will receive a certificate and the learning activity can be added to your RCP CPD diary as a self-certified entry.

This activity will be available for CPD credit for 2 years following its publication date. At that time, it will be reviewed and potentially updated and extended for an additional period.